

Drought Management in India



For Prelims: Drought , State Disaster Response Fund , National Disaster Response Fund , Disaster Management Act, 2005 , El Niño

For Mains: Drought and its types, Challenges of fragmented water governance, Drought and food security and rural livelihoods.

Source: TH

Why in News?

The **Maharashtra Government has declared drought conditions in 265 out of 358 talukas** , covering nearly **74% of the state's landmass** , due to deficient monsoon rainfall. Similar **drought-like conditions and water stress** are being reported in **Karnataka, Telangana, Andhra Pradesh and parts of Rajasthan** , highlighting India's growing vulnerability to **water scarcity and drought** .

- With India supporting nearly **18% of the global population with only 4% of the world's freshwater resources** , drought poses significant risks to **agriculture, food security, rural livelihoods and economic growth** .

Summary

- **Drought in India is a multidimensional challenge** involving rainfall variability, climate change, groundwater depletion and unsustainable agriculture, affecting **food security, livelihoods and water security** .
- India needs to shift from **reactive drought relief to proactive drought-proofing** through **early-warning systems, climate-resilient crops, micro-irrigation, integrated water governance and community-led conservation** .

What is a Drought?

- **About:** Drought is generally considered a **deficiency in rainfall/precipitation over an extended period**, resulting in a water shortage that causes adverse impacts on vegetation, animals, and people.
- **Drought Scenario in India:** According to the **Drought Atlas of India (1901-2020)**, nearly **two-thirds of India is drought-prone**. During this period, about **56% of India's area** experienced moderate to exceptional drought, affecting around **300 million people and 150 million cattle**.
 - Crop losses caused an estimated **USD 8.7 billion economic loss** and reduced **agricultural GDP by 3.1%**.
- **Changing Face of Drought in India:** Traditionally, drought was seen as a phenomenon of arid and semi-arid areas.
 - However, regions with high rainfall now face severe scarcities. For example, **Cherrapunji (Meghalaya)**, one of the **world's highest rainfall areas**, now faces drought for almost nine months of the year, while the western part of **Jaisalmer (Rajasthan)** records around 9 cm of rainfall in a year.
- **Classification of Drought:** The **2009 Manual of Drought Management** issued by the Union Ministry of Agriculture (prepared for Ministry by National Institute of Disaster Management) classifies droughts into three categories:
 - **Meteorological Drought:** It refers to a **deficiency in precipitation compared with the long-term average** over an extended period.
 - It is generally the **first stage of drought** and can trigger other forms of drought. A rainfall deficit of **26-50%** is considered a moderate drought, while a deficit of **more than 50%** is considered a severe drought.
 - **Hydrological Drought:** It occurs when there is a **shortage of surface and groundwater resources** needed for normal and specific uses.
 - It may develop even when rainfall is normal or above normal if **water withdrawal and consumption exceed the available water reserves**, leading to declining river flows, reservoir levels and groundwater availability.
 - **Agricultural Drought:** Occurs when **soil moisture and rainfall are inadequate for crop growth**, causing crop stress and reduced yields.
 - In India, it is defined as **4 consecutive weeks** of severe meteorological drought with **>50% rainfall deficiency or weekly rainfall ≤ 5 cm** from **mid-May to mid-October (Kharif season)**, when **~80% of India's crops are planted**, or **6 consecutive weeks** during the rest of the year.
- **Drought Declaration:** Traditionally, District Collectors recommended drought declaration using the **annewari / paisewari / girdawari system** (estimating agricultural losses post-harvest).
 - However, to facilitate early determination, the **Manual for Drought Management, 2009** recommends four standard monitoring tools.
 - **Rainfall Deficiency:** Drought can be declared if the total rainfall from June-September is less than 75% of the average, or June-July rainfall is less than 50%.
 - **Area under Sowing:** Sowing area of crops is less than 50% of the total cultivable area by the end of July/August.
 - **Normalized Difference Vegetation Index (NDVI):** An index indicating the density of vegetation based on satellite remote sensing (via **NADAMS**). States could declare a drought when the deviation of the NDVI value from the normal is 0.4 or less.
 - **Moisture Adequacy Index (MAI):** Based on a calculation of weekly water balance (ratio of actual to potential evapotranspiration). An MAI below 25 indicates a severe agricultural drought.
- **Regulation:** The **State Government is the final authority** for declaring an area drought-affected. The Union Government has issued two key reference documents: **Manual for Drought Management**,

2009 by the Ministry of Agriculture and **Guidelines for Management of Drought, 2010** by NDMA.

- **Both are non-binding** , allowing states to adopt criteria suited to local irrigation and crop conditions.
- **Financial Relief:** Financial assistance is initially provided through the **State Disaster Response Fund (SDRF)**.
 - If SDRF funds are depleted. The Government of India sends an Inter-Ministerial Central Team to assess losses, after which assistance is released from the **National Disaster Response Fund (NDRF)**.
 - While the Ministry of Home Affairs oversees relief for most natural calamities, the **Ministry of Agriculture and Farmers' Welfare is the nodal ministry for managing droughts, hailstorms, and pest attacks**.
- **Supreme Court's Directives:** In *Swaraj Abhiyan vs Union of India (2016)* , the Supreme Court held that **drought qualifies as a "disaster" under the Disaster Management Act, 2005** .
 - The Court also linked **drought declaration with Article 21** , observing that **failure to declare drought can deprive vulnerable people of essential assistance** required to live with dignity.

What are the Causes of Drought in India?

Meteorological and Climatic Causes:

- **Erratic South-West Monsoon:** India receives roughly 70-80% of its annual rainfall during the four-month monsoon season. Any delay in onset, prolonged "monsoon breaks," or early withdrawal triggers severe **Meteorological Drought** .
- **El Niño Southern Oscillation (ENSO):** The warming of the **equatorial Pacific Ocean (El Niño)** disrupts the **Walker Circulation**, historically correlating with severe rainfall deficits and widespread droughts in India.
- **Climate Change:** Global warming has altered local micro-climates, leading to extreme weather events where short, intense bursts of rain are followed by prolonged dry spells, severely impacting the **Moisture Adequacy Index (MAI)** .

Anthropogenic (Human-Induced) Causes:

- **Skewed Cropping Patterns:** Driven by **Minimum Support Price (MSP) policies** and free electricity, farmers cultivate water-guzzling crops like sugarcane in Marathwada and paddy in Punjab/Haryana (semi-arid zones).
 - This leads to a massive "virtual water export" from water-stressed regions, causing **Agricultural Drought** .
- **Over-exploitation of Groundwater:** India is the **world's largest extractor of groundwater**.
 - Unregulated extraction via deep borewells has decoupled agriculture from annual rainfall recharge, permanently depleting aquifers and triggering **Hydrological Drought** .
- **Ecological Degradation:** Deforestation, destruction of wetlands, and encroachment on traditional water bodies (like **Johads, Baolis , and Eris**) have drastically reduced the natural water-retention capacity of the soil.

What are the Key Challenges in Drought Management in India?

- **Non-Binding Nature of Guidelines:** The **Manual for Drought Management, 2009** and the NDMA's **Guidelines for Management of Drought, 2010** are merely advisory.
 - The flexibility available to states can sometimes lead to variations in the timing and methodology of drought declaration.
- **Delayed Declaration & Relief:** Many states still rely on post-harvest crop-cutting **experiments (*annewari / paisewari*)** rather than modern early-warning indicators like the **NDVI** .
 - This delays the declaration of early or mid-season droughts, denying farmers timely relief from the **SDRF** .
- **Fragmented Water Governance:** Water management is siloed between the **Central Water Commission** (surface water) and the **Central Ground Water Board** (groundwater), preventing a holistic, basin-level approach to water security.
- **Focus on Relief over Mitigation:** Despite the **Disaster Management Act, 2005** mandating proactive risk reduction, state machinery largely activates only for reactive crisis management (distributing tankers and fodder) rather than long-term drought-proofing.
- **Low Penetration of Micro-Irrigation:** Despite schemes like PMKSY, the adoption of drip and sprinkler irrigation remains highly concentrated in a few states due to high initial capital costs for marginal farmers.

What Measures are Needed to Strengthen Drought Management in India?

- **Agronomic Reforms:**
 - **Agro-Climatic Zoning:** Strictly aligning cropping patterns with regional water availability.
 - **Crop Diversification:** Incentivizing a shift from paddy and sugarcane to drought-resistant crops like **Millets (*Shree Anna*)** , pulses, and oilseeds through robust MSP and procurement support.
- **Technological and Scientific Interventions:** State governments must be mandated to use satellite data from the **National Agricultural Drought Assessment and Monitoring System (NADAMS)** , enforcing drought declarations based on objective deviations in **NDVI** and **MAI** .
 - Aggressively scaling up the *Per Drop More Crop* component of the **Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)** .
- **Institutional Restructuring:** Implement the **Mihir Shah Committee's (2016)** recommendation to create a **unified National Water Commission** for integrated surface and groundwater management.
- **Decentralized Groundwater Budgeting:** Empowering Gram Panchayats to audit and regulate village-level groundwater extraction, scaling up models like the Atal Bhujal Yojana.
- **Rejuvenating Traditional Water Bodies:** Leveraging **VB-GRAM G Act, 2025** funds and the **Jal Shakti Abhiyan** to de-silt tanks, build check dams, and construct rainwater harvesting structures.

Conclusion

Drought in India is a **governance, agricultural and water-security challenge** , not merely a rainfall deficit. India must shift from **reactive drought relief to proactive drought-proofing** through early-warning systems, climate-resilient crops, efficient irrigation and community-led water conservation. This is essential for **livelihood security, food security and sustainable economic growth** .

Drishti Mains Question:

Drought in India is increasingly a governance and water-management challenge rather than merely a rainfall-deficit phenomenon. Discuss.

Frequently Asked Questions (FAQs)

1. What are the three major types of drought?

Meteorological drought results from rainfall deficiency, **hydrological drought** from inadequate surface and groundwater, and **agricultural drought** from insufficient soil moisture for crops.

2. Who has the final authority to declare drought in India?

The **State Government** is the final authority for declaring an area drought-affected, using indicators such as **rainfall, sowing area, NDVI and MAI** .

3. What is the role of NADAMS in drought management?

NADAMS uses **satellite-based remote sensing and NDVI** to monitor agricultural drought and support objective and timely drought assessment.

4. How is drought relief financed in India?

Initial financial assistance is provided through the **State Disaster Response Fund (SDRF)** . When additional assistance is required, support may be provided from the **National Disaster Response Fund (NDRF)** after assessment.

5. What did the Supreme Court establish in *Swaraj Abhiyan vs Union of India (2016)*?

The Supreme Court recognised **drought as a disaster under the Disaster Management Act, 2005** and linked timely drought declaration and relief with the **Article 21 right to life and dignity** .

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/z_ILegm9oAg](https://www.youtube.com/embed/z_ILegm9oAg)]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Mains

Q. Discuss the recent measures initiated in disaster management by the Government of India departing from the earlier reactive approach. **(2020)**

Q. Drought has been recognized as a disaster in view of its spatial expanse, temporal duration, slow onset, and lasting effects on vulnerable sections. With a focus on the September 2010 guidelines from the National Disaster Management Authority (NDMA), discuss the mechanisms for preparedness to deal with likely El Nino and La Nina fallouts in India. **(2014)**

Need to Revisit India's Nuclear Doctrine



For Prelims: Pokhran-II , Kargil War , Mutually Assured Destruction (MAD) , Nuclear Command Authority , Line of Actual Control , Artificial Intelligence , Rooppur Nuclear Power Plant .

For Mains: India's Nuclear Doctrine and Strategic Autonomy, No First Use (NFU) and Credible Minimum Deterrence

Source: TH

Why in News?

India's nuclear doctrine , largely unchanged since its operationalisation in 2003, is facing renewed scrutiny amid evolving strategic, technological and geopolitical challenges.

- The **Bangladesh-US Memorandum of Understanding (MoU)** on strategic civil nuclear cooperation signed in September 2026 is part of the **changing regional security environment**, prompting debate on whether **India's nuclear doctrine should be revisited** without being completely overhauled.

Summary

- India's nuclear doctrine, based on No First Use (NFU), Credible Minimum Deterrence (CMD) and Massive Retaliation, faces new challenges from China-Pakistan nuclear convergence, Tactical Nuclear Weapons (TNWs) and emerging technologies.
- India needs periodic doctrinal review, a stronger nuclear triad and second-strike capability, resilient NC3 systems, and flexible response options to maintain credible deterrence while preserving strategic stability.

What is India's Current Nuclear Doctrine?

- **India's Nuclear Doctrine:** India's nuclear doctrine emerged in the late 1990s. The **1998 Pokhran-II tests** established **India as a nuclear weapons state** , while the **1999 Kargil War** demonstrated that nuclear deterrence did not rule out conventional conflicts.
 - The **Cabinet Committee on Security (CCS)** operationalised the doctrine in 2003, based on the 1999 Draft Nuclear Doctrine. The doctrine rests on the pillars of "**credible minimum deterrence**" and a declared "**No First Use**" (NFU) posture.
- **Core Principles of India's Nuclear Doctrine:**
 - **Credible Minimum Deterrence:** Maintaining a sufficient, non-country-specific arsenal to deter adversaries by guaranteeing unacceptable damage.

- **No First Use (NFU):** Employing nuclear weapons *only* in retaliation to a nuclear attack on Indian territory or forces.
- **Massive Retaliation:** Ensuring any retaliatory strike is "massive" enough to inflict incalculable damage, reinforcing **Mutually Assured Destruction (MAD)**.
- **Strict Civilian Political Control:** Authorization rests solely with the **Nuclear Command Authority (NCA)** (comprises a Political Council and an Executive Council):
 - **Political Council** (Chaired by the Prime Minister): Sole authority to release nuclear weapons.
 - **Executive Council** (Chaired by the **National Security Advisor**): Provides strategic inputs and executes directives.
- **Negative Security Assurance & Exceptions:** Pledging not to use nuclear weapons against **non-nuclear weapon states**, *except* retaining the option to retaliate against major **Chemical or Biological weapon attacks**.
- **Commitment to Disarmament:** Maintaining a moratorium on nuclear testing and advocating for verifiable, non-discriminatory global nuclear disarmament.
- **India's Nuclear Arsenal:** According to the **SIPRI Yearbook 2026**, India's nuclear arsenal expanded to an estimated **190 warheads** (12 deployed, 178 in reserve), positioning it slightly ahead of Pakistan (170) but well behind China (620).
- Furthermore, the report highlighted a significant shift in regional deterrence during the May 2025 **"Operation Sindoor"** confrontation, marking the first time India and Pakistan actively integrated cyber warfare into a military conflict.

What is the Need to "Revisit" India's Nuclear Doctrine?

- **China-Pakistan Strategic Convergence:** When the doctrine was drafted, China and Pakistan presented **distinct, isolated threats**.
 - Today, their military and nuclear postures are deeply interconnected. China is rapidly expanding its nuclear arsenal and modernizing infrastructure along the **Line of Actual Control (LAC)**, while Pakistan relies heavily on nuclear posturing to offset India's conventional military superiority.
- **The "Massive Retaliation" Dilemma:** Pakistan has developed **low-yield Tactical Nuclear Weapons (TNWs)**, such as the **Nasr missile**, that could be used in a limited conflict with India to offset India's conventional military advantage.
 - This creates a challenge for India's **Massive Retaliation doctrine**, which promises a large-scale nuclear response to a nuclear attack.
 - Experts therefore debate whether India should retain **Credible Minimum Deterrence (CMD) and No First Use (NFU)** while developing **more flexible and proportionate nuclear response options**.
- **Disruptive Technologies & NC3 Vulnerability:** The advent of **Artificial Intelligence (AI)**, cyber warfare, anti-satellite (ASAT) weapons, and **Hypersonic Glide Vehicles (HGVs)** severely compresses early-warning and decision-making timelines.
 - A review is essential to harden India's **Nuclear Command, Control, and Communications (NC3)** infrastructure against cyber-sabotage, ensuring the survivability of second-strike capabilities.
- **Great Power Competition in the Neighbourhood:** Nuclear dynamics in South Asia are increasingly multipolar. The September 2026 **Bangladesh-U.S. Strategic Civil Nuclear MoU** perfectly illustrates this shift.

- While focused on peaceful energy diversification, the US's entry into Dhaka's nuclear sector, which is historically dominated by Russia (via the **Rooppur Nuclear Power Plant**, where India participates by supplying non-critical equipment, materials, and training through a trilateral framework) and courted by China, highlights the intense geopolitical scramble in the Indo-Pacific.
- For India, this implies its nuclear diplomacy and strategic autonomy must navigate a congested neighbourhood where **external superpowers are actively leveraging civil nuclear technology** to build influence right on its borders.

What Measures are Needed to Strengthen India's Nuclear Doctrine?

- **Enhancing Nuclear Triad:** For the **NFU** pledge to remain credible, India must achieve **Continuous At-Sea Deterrence (CASD)** .
 - This requires accelerating the commissioning of nuclear-powered ballistic missile submarines (**SSBNs like INS Arighat and subsequent S4 class**) and fielding **Multiple Independently Targetable Reentry Vehicles (MIRV)** demonstrated in the Agni-V Divyastra test to overwhelm **adversary Ballistic Missile Defence (BMD) systems**.
- **Hardening C4ISR and Space Resilience:** Modernizing the **Nuclear Command, Control, and Communications (NC3)** network is critical.
 - India must secure its early-warning architectures against gray-zone threats, including **cyber-sabotage, AI-driven data spoofing, and anti-satellite (ASAT) blinding**.
 - Robust space-based surveillance guarantees the **Nuclear Command Authority (NCA)** receives uninterrupted intelligence during crisis timelines.
- **Implementing Strategic Committee Reforms:** Building on the recommendations of the **Naresh Chandra Task Force on National Security (2012)** and the **Kargil Review Committee (1999)**, the integration of the **Chief of Defence Staff (CDS)** into the nuclear decision-making matrix must be optimized.
 - The CDS acts as the vital bridge between the civilian Political Council and the tri-services **Strategic Forces Command (SFC)** , ensuring joint operational readiness.
- **Cultivating Strategic Ambiguity:** To address the tactical nuclear weapon dilemma without altering the NFU text, India should cultivate calculated strategic ambiguity. Developing flexible, proportional response options ensures adversaries cannot exploit predictable gaps on the nuclear escalation ladder.

Conclusion

India's **nuclear doctrine** has ensured **strategic stability, credible minimum deterrence and policy continuity** for over two decades. Rather than a complete overhaul, **periodic review, survivable nuclear forces and secure command systems** can help India maintain effective deterrence amid evolving geopolitical and technological challenges.

Drishti Mains Question:

"India's nuclear doctrine must balance strategic stability with the changing nature of nuclear threats." Discuss in the context of China-Pakistan nuclear convergence.

Frequently Asked Questions (FAQs)

1. What are the key pillars of India's nuclear doctrine?

Credible Minimum Deterrence, No First Use, Massive Retaliation, civilian political control and commitment to nuclear disarmament are its major principles.

2. What is No First Use (NFU) in India's nuclear doctrine?

NFU means India has declared that it will **not use nuclear weapons first**, but will retain the right to retaliate against a nuclear attack.

3. What is Credible Minimum Deterrence (CMD)?

CMD means maintaining a **sufficient and survivable nuclear capability** to deter adversaries by assuring unacceptable damage, without pursuing an unlimited arsenal.

4. Why do Tactical Nuclear Weapons pose a challenge to India?

Low-yield **TNWs** are designed for limited battlefield use, creating questions about how India's **Massive Retaliation** doctrine would respond to a limited nuclear strike.

5. How can India strengthen its nuclear deterrence?

India can strengthen **nuclear survivability, the nuclear triad, SSBN capability, NC3 networks, space-based surveillance and cyber resilience**, while periodically reviewing its doctrine.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/wiEzt2PuKAK](https://www.youtube.com/embed/wiEzt2PuKAK)]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. In India, why are some nuclear reactors kept under "IAEA safeguards" while others are not? (2020)

- (a) Some use uranium and others use thorium
- (b) Some use imported uranium and others use domestic supplies
- (c) Some are operated by foreign enterprises and others are operated by domestic enterprises
- (d) Some are State-owned and others are privately owned

Ans: (b)

Mains

Q. With growing energy needs, should India keep on expanding its nuclear energy programme? Discuss the facts and fears associated with nuclear energy. (2018)

The 'Enemy State' Clauses in the UN Charter



Source: IE

Japan's Prime Minister at the **United Nations General Assembly (UNGA)** urged the UN to eliminate the **"enemy state" clauses** from the **UN Charter**.

- **Background:** The UN Charter was **drafted towards the end of World War II (WWII)** and came into force in **October 1945**.
 - The victorious **Allied Powers** (including the US, UK, Soviet Union, and China) inserted mechanisms to prevent the **resurgence of militarism or fascism in the defeated Axis Powers (mainly Japan, Germany, Italy, and their allies)**.
 - The enemy state clauses are an exception to the post-1945 principle under Article 2(4) of the UN Charter, which generally prohibits the threat or use of force against states.
- **Key Articles Pertaining to the Clauses:** The term "enemy state" is explicitly mentioned in three articles of the UN Charter: **Articles 53, 77, and 107**.
 - **Article 53:** Allows regional arrangements or agencies to take enforcement action against an "enemy state" **without the prior authorization of the UN Security Council (UNSC)**.
 - **Article 53(2)** defines an enemy state as any nation that was an enemy of a signatory of the **UN Charter during WWII**.
 - **Article 77:** Pertains to the **UN "trusteeship system"** and mentions territories detached from "enemy states" as a result of WWII.
 - **Article 107:** Validates and protects actions taken by Allied governments against the Axis powers during and immediately after the war.
- **Japan's Stance on Obsolescence:** Japan joined the United Nations in 1956, after post-war reconstruction and reconciliation with the international community.
 - Japan argues these clauses no longer reflect modern geopolitical realities. The UNGA previously recognized these clauses as "obsolete" in 1995 and adopted a resolution supporting their deletion in 2005.
- **Geopolitical Weaponization:** The outdated clauses are currently being invoked by China and Russia as diplomatic leverage to criticize **Japan's increased defense spending and to invalidate** Japan's territorial claims over the southern **Kuril Islands**.
- **Hurdles to Amendment:** Amendment requires adoption by a two-thirds majority of the UN General Assembly, and ratification by two-thirds of UN member states, including all **permanent members of the Security Council who hold veto power**.
 - They hesitate to open the Charter for revision, fearing it could trigger broader demands for **UNSC structural reforms and membership expansion**, which Japan strongly advocates.
- **Significance for India:** Japan's concerns over outdated UN Charter provisions reinforce India's long-standing call for UN reform that reflects 21st-century geopolitical realities.
 - As a **G4 member with Japan, Germany and Brazil**, **India** supports a more representative, legitimate and effective multilateral system.

Read more: [Debate on UNSC Veto Power](#)

Steel Industry Safety Council



Source: PIB

The **Ministry of Steel** has decided to establish the **Steel Industry Safety Council (SISC)** as an **apex-level industry safety body** to strengthen safety standards, accident prevention and safety culture across India's steel industry.

- **About SISC:** The Steel Industry Safety Council (SISC) will function under the administrative control of the **Ministry of Steel** and will focus on improving safety practices, developing standards and reducing industrial accidents.
- **Composition:** The Council will be headed by the **Secretary, Ministry of Steel**, as Chairman and will include senior officials, CEOs of integrated steel producers, and representatives from key technical and statutory organisations such as the **Directorate General of Mines Safety (DGMS)**, **Petroleum and Explosives Safety Organisation (PESO)**, **National Disaster Management Authority (NDMA)** and recognised steel technology institutions.
- **Functions of SISC:** It will assess safety conditions in the steel industry, conduct studies, approve safety standards and recommended practices, and review serious accidents and near-miss incidents.
- **Steel Industry Safety Directorate (SISD):** A technical support body, SISD, will be established to **provide expertise in process safety**, blast furnaces, steel melting, rolling mills, coke ovens, electrical and mechanical safety, occupational health, fire engineering and disaster management.
- **Role of SISD:** It will implement Council decisions, **conduct safety audits, investigate serious incidents**, maintain accident and near-miss databases, review emergency preparedness and strengthen safety training programmes.
- **Safety Standards Development:** SISD will prepare and issue **Steel Industry Safety Standards, Recommended Practices and Guidelines** suited to Indian conditions to ensure uniform safety practices across the sector.
- **Significance:** The establishment of SISC and SISD aims to create a stronger safety culture, improve industrial accident prevention and promote consistent safety management practices in India's steel industry.

Read more: [India's Steel Sector](#)

India Reaffirms Commitment to Cetacean Conservation at IWC-70



Source: PIB

India has reaffirmed its commitment to the **conservation of whales and other cetaceans** at the **70 th meeting of the International Whaling Commission (IWC-70)** held in **Hobart, Australia** .

- **India's Stand:** India emphasized that cetacean conservation requires science-based approaches, international cooperation and practical measures to address threats faced by whales, dolphins and other related species.
- **Importance of Scientific Research:** India highlighted the need for improved scientific information on cetacean **population status, distribution, movements** , breeding ecology and threats to support informed conservation decisions.
- **International Cooperation:** India stressed collaboration among **range countries** , scientific institutions, governments, researchers and local communities to develop effective and locally suitable conservation strategies.
- **International Whaling Commission (IWC):** The IWC was established under the **International Convention for the Regulation of Whaling** , signed in Washington, D.C., on **2nd December 1946** , and serves as the global body for the management of whaling and the conservation of whales.
- **IWC-70 Significance:** The 70th meeting coincides with the **80th anniversary of the signing of the International Convention for the Regulation of Whaling** . The IWC currently has **89 member governments** , and India has been a member since **1981** .
- **India's Conservation Measures:** Cetaceans in Indian waters receive the highest legal protection under the **Wild Life (Protection) Act, 1972** . India has also established **Marine Protected Areas** and launched **Project Dolphin** for conservation of dolphins, whales, porpoises and aquatic ecosystems.
- **Focus Species and Future Approach:** India has given special importance to the conservation of the **Arabian Sea humpback whale** , identified among 25 species for focused conservation efforts, while supporting future IWC work based on science, cooperation and practical conservation action.

CETACEANS

Whales, Dolphins and Porpoises

- Cetaceans are marine mammals that include whales, dolphins and porpoises.
- They are completely aquatic, meaning they spend all their time in the ocean and cannot survive on land.
- They have two front flippers and a tail with two horizontal extensions (called flukes) that provide powerful swimming ability.



TYPES OF CETACEANS

Cetaceans are separated into two groups: Toothed whales and Baleen whales.

Toothed Whales (Odontocetes)

- Have teeth
- One blowhole (opening)
- Over 73 species (including dolphins, porpoises, sperm whales, beaked whales, belugas, narwhals and even freshwater dolphins)
- Size range: 5 feet (1.5 m) vaquita to 60 feet (21.1 m) sperm whale
- Usually social and live in groups
- Use echolocation (like bats) to detect objects – they produce sounds which bounce back as an echo and give them a "picture" of their surroundings.



Baleen Whales (Mysticetes)

- Do not have teeth
- Have baleen plates (keratin) to filter small marine organisms from water
- Two blowholes
- Includes large whales like blue whale, humpback whale and gray whale
- Generally larger in size than toothed whales



EXAMPLES OF CETACEANS



Common Bottlenose Dolphin (Dolphin)



Gray Whale (Whale)



Harbor Porpoise (Porpoise)



Humpback Whale (Whale)



Blue Whale (Whale)



Vaquita (Porpoise)

EVOLUTION AND ADAPTATIONS

- 1 Land ancestors**

Millions of years ago, the ancestors of whales lived on land. They probably looked like small dogs, but were more closely related to hippos. They went into the ocean about 60 million years ago.


- 2 Transition to ocean**

Over time, these ancestors gradually adapted to live in water and eventually became fully aquatic.


- 3 Key physical adaptations**

Skull elongated and nostrils shifted to top of head (blowholes)

Lost the back legs

Front legs turned into paddle-shaped flippers

Tail widened to form flukes

Thick layer of fat (blubber) to keep warm


- 4 Adaptations for diving**

Ability to store more oxygen in their blood and muscles

More blood volume relative to their body size than land mammals

Specialised physiology to withstand deep diving



POPULATION STATUS AND THREATS

- Many whales are endangered, largely due to past hunting.
- In the past, people used whale products for various purposes:



Blubber oil for lamps and other items



Whale meat for food or pet food



Sinews for tennis racquet strings



Baleen as supports in women's underwear



Ambergris (from sperm whale) used in perfumes – very valuable



Read more: [Project Dolphin](#)